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The Editor-in-Chief
*Computers in Biology and Medicine*
Elsevier

**Re: Submission of an original research article,** "Predicting breast cancer grade from routine clinical data: an open multi-cohort study with transport to Saudi patients"

Dear Editor-in-Chief,

I am pleased to submit the above manuscript for consideration as a full-length research article in *Computers in Biology and Medicine*.

Histological grade is one of the strongest prognostic factors in breast cancer, yet it is assigned only after tissue examination and is only moderately reproducible between pathologists. Machine-learning studies increasingly try to predict grade from images, but there is no transparent, externally validated baseline showing how much grade the routine clinical record already carries. Such studies are also exposed to data leakage, because public breast cancer datasets contain variables that encode grade. Finally, it is unknown whether models developed on Western data give correct predictions for patients in Saudi Arabia, where breast cancer presents younger and is more often hormone-receptor-negative.

To address these gaps, we built an open, checksummed data model of 5025 grade-labelled invasive breast cancers from 7 public cohorts. We then compared seven models, from a prevalence baseline to gradient boosting, with nested cross-validation and frozen validation in five external cohorts on three continents. The main findings are:

- Routine variables predict grade moderately and consistently (macro-AUC 0.72–0.73 internally; 0.64–0.80 externally). Gradient boosting did not improve on logistic regression (difference +0.001, 95% CI -0.006 to +0.008).
- Predictor effects transported across cohorts, but calibration did not: the same model under-predicted grade 3 in one cohort and over-predicted it in another. A simple intercept update corrected both.
- Common leakage shortcuts inflated apparent performance by up to +0.297 in macro-AUC, more than the gap between a useless and the best correctly validated model.
- In 637 patients with breast MRI, 529 radiomic features did not improve on the routine clinical variables.
- Re-weighting the cohorts to published Saudi case-mix showed that Western-trained models would over-predict high-grade disease in Saudi patients, so local recalibration is essential.

We believe the work fits the journal's scope in computational methods for medicine, and it offers four things that readers will find useful. It provides a reproducible benchmark that image-based grading tools can be measured against. It shows that a simple, interpretable model is sufficient. It gives concrete evidence of how leakage inflates results. And it offers practical guidance on recalibrating models before they are used in a new population.

The study is reported according to TRIPOD+AI (prediction) and STROBE (descriptive analyses); the completed TRIPOD+AI checklist is included in the supplementary material. It uses only de-identified, publicly released data, so no ethics approval was required. All code, configuration files and results are available in an anonymised repository [insert anonymous repository link]. A single command reproduces every table, figure and number in the manuscript.

I confirm that this manuscript is original, has not been published previously, and is not under consideration

for publication elsewhere. [All authors have approved the submission.] The author declares no competing interests and received no specific funding for this work [confirm]. The use of AI-assisted tools is disclosed in the manuscript, as required by Elsevier policy. Because the manuscript has been prepared for double-anonymised review, author details are provided only in this letter and on the separate title page.

[Optional: suggested reviewers with name, affiliation, e-mail and a one-line reason for each, chosen from researchers with no conflict of interest, for example experts in clinical prediction-model validation or computational breast pathology.]

Thank you for considering this manuscript. I look forward to hearing from you.

Yours sincerely,

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